

ANTHONY BWEMBYA

COMPUTER VISION | OPTICAL SENSOR DEVELOPMENT & CALIBRATION | AI/ML

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CORE EXPERTISE

- Optical and imaging sensor calibration
- Computer-vision-oriented data analysis
- Signal and image-quality assessment
- Sensor-response modelling and validation
- AI/ML for physical measurements
- Feature extraction and anomaly detection
- Multisensor synchronisation and fusion
- Uncertainty quantification
- Hardware-software integration

PROGRAMMING & DATA

- Python, C++, Bash, SQL
- NumPy, SciPy, pandas, Matplotlib
- Scientific image/data processing
- Optimisation and statistical inference
- Linux, Git, Jupyter
- HPC and SLURM workflows
- Automated quality-control pipelines

EDUCATION

PhD, Experimental Physics

Radboud University | 2021–2026

Thesis completed; defence scheduled 2026

MSc, Nuclear Physics & Technology

National Research Nuclear University MEPhI | 2019–2021

BSc, Physics

University of Zambia | 2017

PROFESSIONAL PROFILE

PhD-trained experimental physicist and sensor-data specialist with experience developing, calibrating and validating complex optical, radio and detector-based measurement systems. Strong background in Python/C++ scientific software, signal reconstruction, sensor-response modelling, uncertainty analysis, automated quality control and AI/ML applied to physical measurements. Hands-on exposure to LiDAR, fluorescence cameras, PMTs, detector electronics and low-noise instrumentation. Well positioned for computer vision and optical-sensor R&D roles requiring rigorous calibration, image/signal interpretation, hardware-software integration and reliable model validation.

TARGETED VALUE

- Transforms noisy multidimensional sensor outputs into calibrated, quality-controlled physical measurements using reproducible Python/C++ workflows.
- Applies geometric reconstruction, feature extraction, optimisation and uncertainty propagation—methods transferable to optical imaging and computer-vision pipelines.
- Develops diagnostics that distinguish hardware faults, calibration drift, environmental effects and algorithmic errors.
- Combines experimental instrumentation with data-driven methods from sensor characterisation through model evaluation and validation.

PROFESSIONAL EXPERIENCE

Sensor Development, Calibration and Intelligent Data Processing

Radboud University / Pierre Auger Observatory |
Netherlands / Argentina | 2021–2026

- Developed precision calibration and synchronisation methods for a distributed sensor array, achieving nanosecond-level timing alignment across autonomous stations.
- Built Python and C++ pipelines for waveform processing, spectral analysis, feature extraction, reconstruction, sensor diagnostics and automated quality control.
- Designed algorithms to identify anomalous sensors, ambiguous solutions, timing drift, geometry errors and inconsistent measurements using network-level consistency tests.
- Modelled wave propagation, detector geometry, atmospheric refractivity and sensor response to validate reconstructed events against physical expectations.
- Developed coherent interferometric reconstruction methods for weak signals, culminating in the first successful reconstruction of a highly inclined event using AugerPrime radio-detector data.
- Integrated measured background noise into simulation workflows to evaluate algorithm robustness under realistic conditions.
- Supported LiDAR and fluorescence-detector calibration and operations, including optical/atmospheric monitoring and detector-performance assessment.

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OPTICAL & SENSOR SYSTEMS

- LiDAR and atmospheric monitoring
- Fluorescence-camera operations
- PMT and optical-detector calibration
- RF/radio detector arrays
- DAQ and detector electronics
- VNA, AWG, DAC and RF/DC chains
- Low-noise measurements
- Field and laboratory validation

AI / ML

- Supervised and unsupervised learning concepts
- Classification, regression and clustering
- Feature engineering and model validation
- Anomaly detection and quality control
- Uncertainty-aware interpretation
- Teaching support in ML for physics

LANGUAGES

- English – fluent
- Russian – good
- Dutch – basic/developing
- Bemba / Nyanja – native

ADDITIONAL

- No visa sponsorship required in the Netherlands
- Available for laboratory, R&D and applied AI roles

ADDITIONAL EXPERIENCE

Experimental Instrumentation & Optical Sensor Experience

Selected laboratory and field work

- Supported LiDAR operations at the Pierre Auger Observatory for atmospheric monitoring, data-quality assessment and validation of conditions affecting optical air-shower measurements.
- Participated in fluorescence-telescope operations, including detector-status checks, observing support, monitoring of telescope and camera performance, and review of data-quality indicators.
- Contributed to fluorescence-camera calibration using controlled light sources and detector-response measurements to assess pixel gain, uniformity, stability and calibration quality.
- Linked optical-instrument operation and calibration outputs with environmental and detector data to support reliable event reconstruction and uncertainty-aware analysis.

Research & Technical Officer – Radiation Measurements and Instrumentation

Radiation Protection Authority, Zambia | 2021–2022

- Performed radiation and spectroscopic measurements, instrument checks, detector calibration, sample preparation, spectral interpretation and technical reporting.
- Investigated anomalous readings, evaluated uncertainty and supported field inspections, customer-facing testing and compliance documentation.

Teaching Assistant – Machine Learning, Physics & Scientific Computing

Radboud University | 2021–2025

- Supported Machine Learning in Particle Physics and Astrophysics, experimental physics, scientific programming and data interpretation.
- Guided students in model evaluation, feature-based analysis, uncertainty, Python workflows and technical communication.

PROJECTS

AI Transcription and Summarisation Workflows | Zapier

- Designed automated AI workflows in Zapier to process recorded or uploaded content through transcription, summarisation and structured-output stages.
- Connected trigger-based applications and AI services to reduce manual handling and create repeatable end-to-end information-processing workflows.
- Configured prompts and workflow steps to generate concise summaries, action points and reusable text from transcribed material.

RELEVANT METHODS

Sensor calibration • image/signal preprocessing • Fourier and frequency-domain analysis • geometric reconstruction • feature extraction • optimisation • anomaly detection • statistical filtering • model-to-data comparison • uncertainty propagation • simulation-based validation • experiment automation • technical documentation